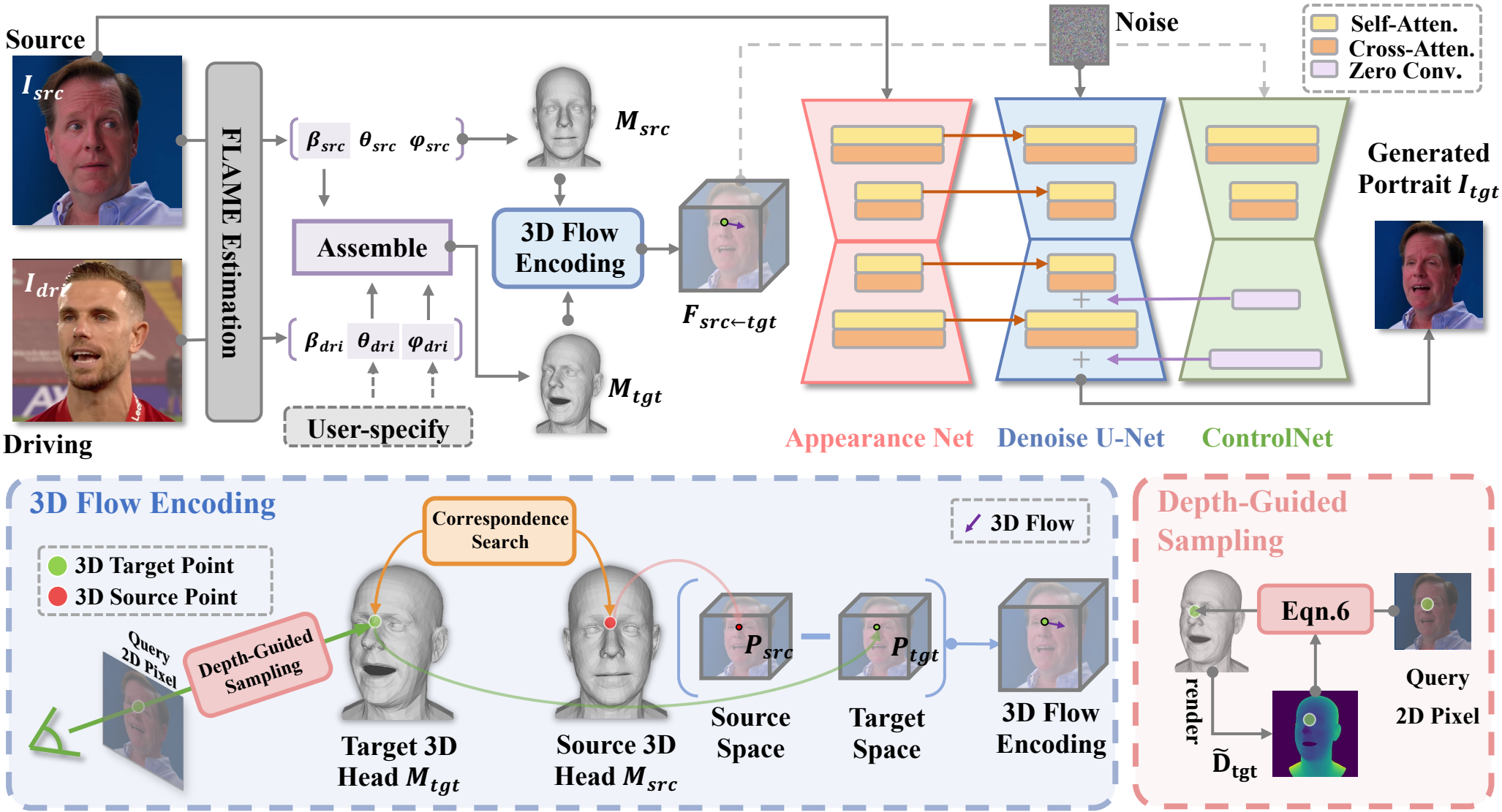
Source
I_{src}
I_{dri}
Driving
FLAME Estimation
β_{src} θ_{src} φ_{src}
β_{dri} θ_{dri} φ_{dri}
Assemble
User-specify
M_{src}
M_{tgt}
3D Flow Encoding
$F_{src \leftarrow tgt}$
Noise
Self-Atten.
Cross-Atten.
Zero Conv.
Generated Portrait I_{tgt}
Appearance Net
Denoise U-Net
ControlNet
3D Flow Encoding
3D Target Point
3D Source Point
Query 2D Pixel
Depth-Guided Sampling
Correspondence Search
Target 3D Head M_{tgt}
Source 3D Head M_{src}
P_{src}
P_{tgt}
Source Space
Target Space
3D Flow
3D Flow Encoding
Depth-Guided Sampling
Eqn.6
render
$\tilde{D}_{tgt}$
Query 2D Pixel